Reconsolidation Theory Leads to Stronger Bias Reduction in LLM Models

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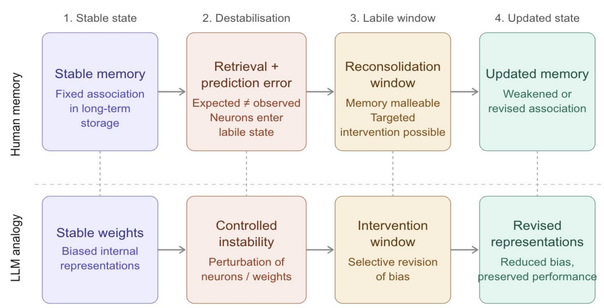
Background

- Large language models can replicate and amplify stereotypes in gender, age, or culture [1].
- Fine-tuning, counterfactual augmentation, and adversarial training reduce bias but require extra data, manual effort, or architecture changes, and treat bias as static rather than dynamic [3].
- Human memory research shows that retrieved memories enter a labile state [2], allowing targeted interventions to update associations without full relearning.
- Introducing controlled, temporary instability in LLM neurons or weights could selectively revise biased patterns while preserving overall model performance.
- This approach targets internal representations directly, reducing reliance on dataset-level modifications.



Methods and Datasets

- Use Sparse Autoencoders (SAEs) to identify which weights are responsible for activating bias-related features in the model's middle layers, and build an Instability Matrix that flags those weights.
- During fine-tuning on a bias intervention dataset, amplify updates to flagged weights by a factor of 1.025 (decaying exponentially over subsequent steps) to simulate a computationally "labile state."
- Compare three conditions, standard fine-tuning, fine-tuning with full instability, and fine-tuning with mild instability, to assess whether the labile state approach reduces bias more effectively than conventional fine-tuning alone.
- Evaluate using the BBQ (Bias Benchmarking for Question Answering) dataset [4], focusing on Age and Gender Identity categories across ambiguous and disambiguated contexts.
- Created a custom age and gender analogies dataset, inspired by the Google Analogy Dataset, using an LLM.
  - This dataset allowed us to evaluate whether induced instability reduces bias while preserving relevant demographic information.

Results

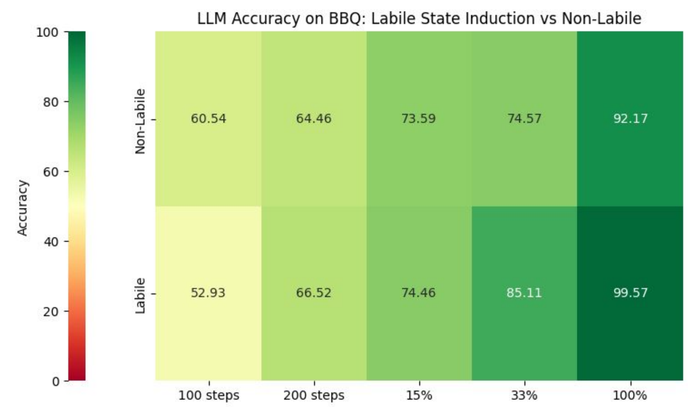


Figure 1. LLM **Accuracy Comparison on BBQ Age Data:** Training with Mild Labile State Induction vs Without

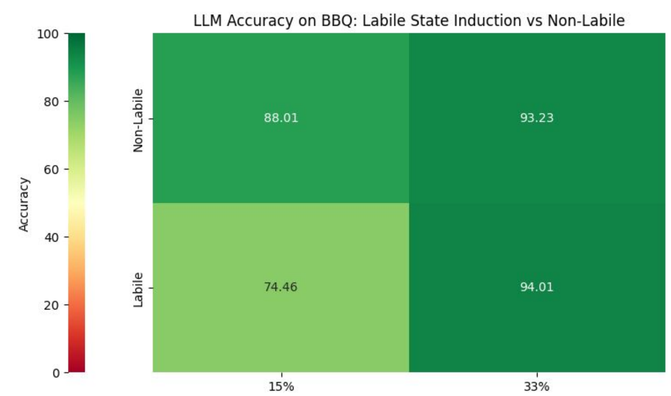


Figure 2. LLM **Accuracy Comparison on BBQ Gender Data:** Training with Mild Labile State Induction vs Without

Conclusions

- We observe a trend in which the Labile State LLM tends to start converging later than the Non-Labile LLM. This is true after both the Age and the Gender Bias dataset training (see Figure 1 and Figure 2)
- We see that eventually, the Labile LLM performs on par with the Non-Labile LLM on test data. Once again, this is true for both training on the Age and the Gender Bias datasets (see Figure 1 and Figure 2).
- At the end, we see for the Age LLMs that the Labile LLM eventually achieves a slight but notably higher accuracy after it completes training. That is, it is able to "unlearn" bias better than the other Non-Labile LLM. Final results are not yet confirmed for the Gender LLM, but we assume the result will be similar: the Labile LLM will outperform the Non-Labile LLM by a small, yet notable, margin.

Discussion & Limitations

- As hypothesized based on principles from Reconsolidation Theory, inducing instability particularly to neurons responsible for biased responses, does improve LLM performance.
- However, the improvement is not particularly large: only a few percentage points.

- We propose the following explanation: Already, when an LLM is training on data, there is some instability induced during batch gradient descent and its variations, whereby the neurons most responsible for "incorrect decisions" (in our case, biased or illogical assumptions about statements included in BBQ) will have their weights adjusted. Our method introduces greater instability to weights that directly result in the bias, but these weights are still adjusted repeatedly through regular finetuning.
- Essentially, one could argue that we are comparing a loose implementation of Reconsolidation Theory present even in the Non-Labile LLMs, with the stronger version present in the Labile LLMs, explaining the close results.

- To address the above limitation, one might propose that in the Non-Labile LLM, we should mitigate the changes to all weights other than those linked specifically to the SAE neuron encoding bias. This would detach the Non-Labile LLM further from embodying the principles of Reconsolidation Theory.

Further Work

- Further compare LLM Accuracy on BBQ Gender Data when training with and without labile state induction.
- Compare the retention of gender and age analogies after training with and without labile state induction, using our custom dataset.
- Determine the exact point at which instability becomes harmful, causing a "collapse" into illogical or repetitive responses.
- Compare the models performance against other debiasing techniques

References

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